

If you are allowed to allocated rate between sensors - not just one bit per sensor (or log m bits like here), you can do much better. The sensors will (distributively) see if their measurement is in a place allowing to distinguish between H_1 and H_2 or not. If not - they will not send (or send less). If so - they will send at high rate. At the limit: only one sensor who go a measurement at a place where f_1 and f_2 differ significantly will send the whole N bits. You will get a better error exponent at the same rate!

DECENTRALIZED DETECTION WITH MANY SENSORS: OPTIMALITY OF EXCHANGEABLE AND IDENTICAL ENCODING POLICIES

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ABSTRACT

We study a class of binary detection problems involving a single fusion center and a large or countably infinite number of sensors. Each sensor acts under a decentralized information structure, accessing only a local noisy observation related to the hypothesis. Based on this observation, sensors select policies to transmit a quantized signal through their actions to the fusion center, which makes the final decision using only these actions. This paper makes the following contributions: i) In the finitely many sensor setting, we provide a formal proof that an optimal encoding policy exists, and such an optimal policy is independent, deterministic, and of threshold type for the sensors and the maximum *a posteriori* probability type for the fusion center; ii) For the finitely many sensor setting, we further show that an optimal encoding policy exhibits an exchangeability (permutation invariance) property; iii) We establish that an optimal encoding policy exists that is symmetric (identical) and independent across sensors in the infinitely many sensor setting under the error exponent cost; iv) Finally, we show that a symmetric optimal policy for the infinite population regime with the error exponent cost is approximately optimal for the large but finite sensor regime under the same cost criterion.

Index Terms— Decentralized detection, large sensor networks, optimality of exchangeable encoding policies, and asymptotic optimality of identical encoding policies.

1. INTRODUCTION

In decentralized detection problems, a collection of sensors gathers local observations from the environment and transmits quantized versions of these observations to a fusion center. The fusion center then decides on one of the hypotheses about the state of the environment based on the quantized observations from the sensors [1, 2]. Numerous studies have been conducted on decentralized detection, addressing both its theoretical and practical aspects; see e.g., [1–5].

The primary objective in a decentralized detection problem is to design the strategies for the sensors and the fusion center to minimize a cost function. Once the strategies at the sensors are determined (fixed), obtaining the optimal decision rule at the fusion center is not challenging, as it reduces to the classical design of optimal decision rules [1, 6–8]. However, when the decision rule at the fusion

center is fixed, determining the optimal decision rules at the sensors generally becomes a challenging problem, mainly because obtaining a globally optimal solution for the joint design of all sensor decision rules typically corresponds to a complex optimization problem. Therefore, in the literature, most studies adopt the person-by-person optimality criterion [1, 2]. In this setting, computing the optimal decision rules at the sensors becomes more manageable as they can be represented by threshold tests [3, 9–18].

Although this approach simplifies the design of sensor decision rules, it may not lead to a globally optimal solution in general [1]. There exists a limited number of studies that focus on global optimality instead of person-by-person optimality for decentralized detection problems [5], [10, Sec. 3], [19, 20, 20, 21]. For example, in a seminal work [21], it is shown that as the number of sensors goes to infinity, an asymptotic solution is to have all the sensors perform an identical likelihood ratio test (LRT) [21].

Our aim in this work is to focus on the global optimization of sensor decision rules in a decentralized binary detection problem and to arrive at the existence and structure results for optimal decision rules considering both finite and infinite numbers of sensors. To achieve this goal, we utilize recent results for large stochastic teams in [22]. In stochastic team decision problems, a finite number of decision-makers (agents) act collectively to minimize a common cost function, but they do not necessarily share their information; see, e.g., [23, 24]. In [22], the optimality of independently randomized symmetric policies has been established for a class of exchangeable stochastic teams. We view the decentralized detection problem as a stochastic team problem involving the sensors and the fusion center, building also on the mathematical machinery developed in [22] while accounting for the unique structure of the detection problem. The main contributions of the paper can be summarized as follows:

1. In Theorem 1, we establish the existence of an optimal policy and that such an optimal policy is deterministic, and of threshold type for sensors in the finite population setting. While optimality of threshold policies (assuming existence) was established in [1, Proposition 2.4], their existence was not proven. Key to our analysis in this case is a reformulation of the problem and the approach via an optimal quantization formulation, considering optimality of threshold policies established in [1].
2. For the finite population setting, in Theorem 2, we show that an optimal encoding policy exists and that such an optimal policy is exchangeable among sensors. In this formulation, the fusion center is allowed to have access to the possible randomness of the encoders' policies.
3. In Theorem 3, we demonstrate that an optimal solution exists and

The research of S. Sanjari and S. Yüksel is supported in part by the Natural Sciences and Engineering Research Council (NSERC) of Canada. S. Gezici acknowledges the support of the Scientific and Technological Research Council of Türkiye (TUBITAK) under the BIDEB 2219 Program. Email address for correspondence: sanjari@rmc.ca

that such an optimal policy is symmetric (identical) and independent among sensors in the infinite population regime under the error exponent cost. Theorem 3 also shows that this symmetric optimal policy for the infinite population regime is near optimal for the large but finite sensor regime.

Due to the space constraint, proofs are not included here (for complete proofs and technical arguments, see [25]).

2. BINARY DETECTION PROBLEM WITH UNIFORM COST ASSIGNMENT

We study a class of binary detection problems in the Bayesian framework with uniform cost assignment (UCA), where the possible hypotheses are denoted by H_1 and H_2 . The true hypothesis H^* can be regarded as a random variable equal to H_i with probability $P(H_i)$ for $i = 1, 2$. We focus on problems with a finite number of sensors as well as a countably infinite number of sensors.

In the finite population setting, there exist N peripheral sensors in the environment, which make observations from the environment with y^i denoting the observation (measurement) of sensor i , taking values in a standard Borel space $(\mathbb{Y}^i, \mathcal{Y}^i)$ for $i \in \{1, \dots, N\}$. Each sensor quantizes its observation via a mapping γ^i that maps the observation y^i into an action u^i . In particular, for a given realization y^i , an action of the sensor i is expressed as $u^i = \gamma^i(y^i) \in \mathbb{U}^i = \{1, \dots, |\mathbb{U}^i|\}$ for $i \in \{1, \dots, N\}$, where $\gamma^i : \mathbb{Y}^i \rightarrow \mathbb{U}^i$ is a policy or decision rule of the sensor i . Each sensor's action is observed by the fusion center (denoted by agent 0), which makes the final decision about the true hypothesis based on the sensors' actions. Therefore, the final decision is stated as $u^0 = \gamma^0(u^1, \dots, u^N)$, where $\gamma^0 : \prod_{i=1}^N \mathbb{U}^i \rightarrow \{1, 2\}$ is the policy (or decision rule) of the fusion center. We use Γ^i to represent the set of all possible policies for agent i for $i = 0, 1, \dots, N$. The collection of policies is denoted by $(\gamma^0, \gamma^1, \dots, \gamma^N)$. The set of all policies is expressed as $\Gamma^0 \times \Gamma^1 \times \dots \times \Gamma^N$. We assume that the action and observation spaces are identical among sensors. We consider the cost function c that depends directly only on the fusion center's decision, u^0 . In particular, we focus on the following uniform cost assignment (UCA):

$$c(H_j, u^0, u^1, \dots, u^N) := c(H_j, u^0) = \begin{cases} 1, & \text{if } u^0 \neq j \\ 0, & \text{if } u^0 = j \end{cases}.$$

The above detection formulation is an instance of a stochastic team problem with the following expected cost:

$$J^N(\gamma^{0:N}) = \mathbb{E}^{\gamma^{0:N}} [c(H, u^{0:N})]. \quad (1)$$

We call the problem involving the cost in (1) as $\mathcal{P}^N$. Our approach is to view the decentralized detection problem, with the fusion rule set as the optimal rule, and thus removed from the system as a static team problem. We characterize the fusion center's optimal decoding rule for any sensor's policy.

For the infinite population problem, following [1, Proposition 5.1], the expected cost (which is the probability of error) goes to zero exponentially as the number of sensors goes to infinity. Hence, we instead study the error exponent as our new cost for the infinite population problem, i.e., the error exponent cost in the finite population for any $\gamma^{0:N}$ is given by

$$J_{EE}^N(\gamma^{0:N}) = \frac{\log(J^N(\gamma^{0:N}))}{N} \quad (2)$$

and the error exponent cost of a policy $\gamma^{0:\infty} := (\gamma^0, \gamma^1, \dots)$ is

$$J_{EE}(\gamma^{0:\infty}) = \limsup_{N \rightarrow \infty} \frac{\log(J^N(\gamma^{0:N}))}{N}. \quad (3)$$

We call the problems involving (2) and (3) by $\mathcal{P}_E^N$ and $\mathcal{P}_E^\infty$, respectively. Next, we introduce our main assumptions.

Assumption 1. Suppose that

- (i) $y^1, \dots, y^N$ are i.i.d., conditioned on H .
- (ii) For any $i = 1, \dots, N$, there exists a probability measure $Q \in \mathcal{P}(\mathbb{Y}^i)$ and a function $f : \mathbb{Y}^i \times \mathbb{H} \rightarrow \mathbb{R}$ such that for any Borel set $\mathbb{A}^i \subseteq \mathbb{Y}^i$,

$$\Pr(y^i \in \mathbb{A}^i | H) = \int_{\mathbb{A}^i} f(y^i, H) Q(dy^i) \quad \forall H \in \mathbb{H}.$$

- (iii) $\mathbb{U}^i = \mathbb{U}$ for $i = 1, \dots, N$, where $\mathbb{U}$ is a finite set.
- (iv) The conditional density $f_{y^i|H}$ of y^i given H exists for every $i \in \{1, \dots, N\}$ and such that the Jacobian matrix of derivatives of $L(y^i)$ given by

$$l^i = L(y^i) := \frac{f_{y^i|H}(y^i|H_2)}{f_{y^i|H}(y^i|H_1)} \quad a.s. \quad (4)$$

is continuous in y^i and nonsingular for all $y^i \in \mathbb{Y}^i$.

By a change of measure argument, Assumptions 1-(i) and 1-(ii) allow us to probabilistically view the observations of sensors as independent of each other and H ; see [10, 26]. Under Assumption 1-(iv), the random variable $l^i := L(y^i)$ admits a density using the inverse function theorem of vector calculus [27, Section 1.11].

Our main goal is to establish the existence and structural results for an optimal solution for the N -sensor setting and its limiting problem with an infinite number of sensors. To this end, we allow joint randomization among the sensors' policies and expand the information of the fusion center to include the realization of joint sensors' policies (common randomness). Thus, we allow the fusion center to have access to the (known common randomization):

$$I_N^{\text{KCR}} := \{u^{1:N}, \gamma^{1:N}\}, \quad (5)$$

under which the common randomness is known to the fusion center. For our analysis, we work with an equivalent information set for the fusion center, which is amenable to our convergence argument as the number of sensors approaches infinity.

3. EQUIVALENT FORMULATION FOR THE FINITELY MANY SENSORS PROBLEM

We first define a class of deterministic threshold policies for sensors adopted from [1].

Definition 1. 1. A deterministic policy for a Sensor ^{i} , γ^i , is a monotone threshold policy if there exist an integer $m^i \in \mathbb{N}$ and threshold constants $\{t_1^i, \dots, t_{m^i}^i\}$ satisfying $0 < t_1^i < t_2^i < \dots < t_{m^i}^i < \infty$ such that

$$\gamma^i(y^i) = d \quad \text{if } l^i \in B_d^i \quad (6)$$

for $B_d^i = [t_{d-1}^i, t_d^i]$ with $d = 2, \dots, m^i - 1$, $B_1^i = [0, t_1^i]$, and $B_{m^i}^i = [t_{m^i-1}^i, \infty]$, where

$$l^i := L(y^i) := \frac{f_{y^i|H}(y^i|H_2)}{f_{y^i|H}(y^i|H_1)} \quad a.s. \quad (7)$$

- 2. A deterministic policy for a Sensor ^{i} , γ^i , is a threshold policy if there exists a permutation mapping τ of $\{1, \dots, m^i\}$ for $m^i \leq |\mathbb{U}^i|$ such that

$$\gamma^i(y^i) = \tau(d) \quad \text{if } l^i \in B_d^i \quad (8)$$

is a monotone threshold policy.

Denote the space of all deterministic threshold policies by Γ_{TS}^N . We note that allowing the number of bins m^i to vary will be useful for establishing compactness properties, which are needed for the existence result. The optimality of threshold policies is stated next.

Lemma 1. [1] Consider $\mathcal{P}^N$. Let Assumption 1 hold. Then,

$$\inf_{\gamma^{1:N} \in \Gamma_N} J^N(\gamma^0, P_\pi) = \inf_{\gamma^{1:N} \in \Gamma_{\text{TS}}^N} J^N(\gamma^0, \gamma^{1:N})$$

for any $\gamma^0 \in \Gamma^0$.

Next, the optimality of the log-likelihood ratio test for the fusion center is stated.

Lemma 2. [1] Consider $\mathcal{P}^N$ under Assumption 1. For any deterministic encoding policy of the sensors, the optimal decoding policy of the fusion center is given by

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) \geq t \\ H_2 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) < t \end{cases} \quad (9)$$

for some $t \in \mathbb{R}$, where $\Delta_N(u^{1:N}; \gamma^{1:N})$ is given by

$$\frac{1}{N} \sum_{i=1}^N \log \frac{\int P^{\gamma^i}(U^i = u^i | l^i) P(dl^i | H_1)}{\int P^{\gamma^i}(U^i = u^i | l^i) P(dl^i | H_2)}. \quad (10)$$

Given the supporting results above, we present an equivalent problem that will be critical for our analysis. For our threshold policies, l^i and Δ_N (given in (7) and (10)) are sufficient statistics for each sensor and the fusion center, respectively. The expected cost $J(\gamma^{0:N})$ can be written as

$$\int \tilde{c}_0(H, \Delta_N) \prod_{i=1}^N \mathbb{I}_{\{\gamma^i(l^i) \in du^i\}} P(dl^i | H) P(dH), \quad (11)$$

where $\tilde{c}_0(H, \Delta_N) := c(H, \gamma^0(\Delta_N(u^{1:N}; \gamma^{1:N})))$ is the new cost function. Hence, following Lemma 1 and Lemma 2, we can reformulate the problem such that each sensor only has access to l^i and the fusion center has access only to the likelihood ratio of Δ_N . Next, we show that an optimal solution exists where sensors' policies are of the threshold type. To this end, we require a topology for the space of threshold encoding policies. We require a strong notion of convergence since the expected cost (11) under the maximum *a posteriori* probability (MAP) decoding rule of the fusion center no longer involves a constant continuous function as the encoding policies vary (see a counterexample in [28, Section 3.1.1] in the context of quantizers). In the following, we endow a topology for threshold policies using that introduced in [28] for quantizers utilizing the total variation distance. To this end, we view each threshold policy γ^i as a stochastic kernel $\gamma^i(k | l^i) = \mathbb{I}_{\{l^i \in B_k^i\}}$ for $k \in \mathbb{U} := \{1, \dots, M\}$, where $B_k^i = \{l^i \in \mathbb{L}^i | t_{k-1} < l^i \leq t_k\}$ are code cells. A sufficient statistics random variable l^i is given by (7). In other words, a threshold policy $\gamma^i \in \mathcal{P}(\mathbb{U} | \mathbb{L}^i)$ is a stochastic kernel, where $\mathbb{L}^i \subseteq \mathbb{R}$ is the set of all possible l^i . Equip $\mathcal{P}(\mathbb{U} | \mathbb{L}^i)$ with the topology induced by the total variation distance $\|\cdot\|_{\text{TV}}$ at any input measure P on $\mathbb{L}^i$, that is, γ_n^i converges to γ^i as n goes to infinity in total variation at input measure P if as $n \rightarrow \infty$

$$\begin{aligned} \|\gamma_n^i - \gamma^i\|_{\text{TV}} &= \sup_{f: \|f\|_\infty \leq 1} \left| \int f(u, l) \gamma_n^i(du | l) P(dl) \right. \\ &\quad \left. - \int f(u, l) \gamma^i(du | l) P(dl) \right| \rightarrow 0. \end{aligned}$$

Theorem 1. Consider $\mathcal{P}^N$. Let Assumption 1 hold. Then, there exists an optimal solution $(\gamma^{0*}, \gamma^{1*}, \dots, \gamma^{N*})$. Furthermore, this policy is a MAP rule for the fusion center and threshold policy for each sensor, i.e., $\gamma^i \in \Gamma_{\text{TS}}^N$ for all $i \in \{1, \dots, N\}$.

The proof of the above theorem follows from the compactness of the space of threshold policies (in [28, Theorem 5.8]) and the continuity of the fusion center's optimization problem in threshold policies for the sensors. Compared to [1], where only the optimality of threshold policies has been discussed, here, we established the existence of an optimal encoding policy, which is of threshold type.

4. OPTIMALITY OF EXCHANGEABLE POLICIES FOR FINITELY MANY SENSORS SETTING

To establish the existence and structural results for an optimal solution to limiting problems with an infinite number of sensors, we need to allow randomization among sensors' policies, which involves various relaxations given the decentralized information structure.

4.1. Randomized policies for sensors

Under Assumption 1, we define randomized policies for each sensor i separately and identify it by Γ^i via the set of probability measures on $\mathbb{U}^i \times \mathbb{L}^i$ with the fixed marginal P on $\mathbb{L}^i$, i.e.,

$$\Gamma^i := \left\{ \gamma^i \in \mathcal{P}(\mathbb{U}^i \times \mathbb{Y}^i) \mid \gamma^i(\mathbb{U}^i, \cdot) = P(\cdot) \right\},$$

which is Borel measurable under the weak convergence topology by [23, 29]. The sequence of randomized policies $\{\gamma_n^i\}_{n \geq 0}$ converges to a policy γ^i if and only if

$$\gamma_n^i(du^i, dl^i) := \pi_n^i(du^i | l^i) P(dl^i) \rightarrow \gamma^i(du^i, dl^i)$$

weakly as $n \rightarrow \infty$ for $\gamma^i(du^i, dl^i) := \pi^i(du^i | l^i) P(dl^i)$. Given this, we define a joint randomized policy $P_\pi \in L^N$ for sensors with L^N as the set of all joint distributions of $\gamma^{1:N}$. Also, a subset L_{EX}^N denotes a set of all joint distributions that are permutation invariant for all permutations of $\{1, \dots, N\}$. Denote the set of all joint distributions that are independent, conditioned on common randomness of z by L_{CO}^N , and the set of all joint distributions that are identical and independent by $L_{\text{PR,SYM}}^N$.

For our analysis, we work with the information set for the fusion center, which is amenable to our convergence argument as $N \rightarrow \infty$. For this information set, we allow the fusion center to have access to

$$I_N^{\text{FS}} = \{u^{1:N}, \mu^N, I_1, \dots, I_N\}, \quad (12)$$

where μ^N is the empirical measure of random policies $\gamma^{1:N}$ and $I_1, \dots, I_N$ (with $I_i \in \{1, \dots, N\}$ valued) are the realized permutations by observing the realization of indices. This allows the fusion center to infer the realized $\gamma^{1:N}$ for any $P_\pi \in L_{\text{CO}}^N$.

4.2. Optimality of exchangeable policies

We now present structural and existence results for optimal encoding policies. Recall that Lemma 2 guarantees that the fusion center policy is a MAP rule, and hence, without any loss of optimality, we can restrict our search for an optimal decoding policy to this class of policies under the information (12). For any randomized encoding policy of the sensors $P_\pi \in L^N = L_{\text{CO}}^N$, γ^0 will be of the form (9), where $\Delta_N(u^{1:N}; \gamma^{1:N})$ in (9) is replaced by

$$\bar{\Delta}_N(u^{1:N}, P_\pi, \mu^N, I_1, \dots, I_N) \quad (13)$$

$$= \frac{1}{N} \sum_{i=1}^N \log \frac{\int \nu(U^i = u^i | l^i, P_\pi, \mu^N, I_1, \dots, I_N) P(dl^i | H_1)}{\int \nu(U^i = u^i | l^i, P_\pi, \mu^N, I_1, \dots, I_N) P(dl^i | H_2)},$$

where

$$\nu(U^i = u^i | l^i, P_\pi, \mu^N, I_1, \dots, I_N) = \gamma^{I_i}(U^i = u^i | l^i) \quad (14)$$

for the realized policy γ^{I_i} of the sensor i .

Theorem 2. Consider $\mathcal{P}^N$ under Assumption 1. Then, there exists an optimal solution (γ^{0*}, P_π^*) , that is exchangeable among sensors with γ^{0*} of the form (9) with (13), i.e., $P_\pi^* \in L_{\text{EX}}^N$.

We now explain the reasoning behind the proof. Since $y^{1:N}$ are i.i.d. conditioned on the hypothesis, the ratios l^i are also conditionally i.i.d. Using this together with the fact that μ^N is permutation invariant, we show that any permutation of the encoding policy does not change the performance, then by convexity of the set randomized encoding policy, we construct an exchangeable policy as an average of all its permutations. Since the realized policies can be deduced from μ^N and $I_1, \dots, I_N$, an exchangeable policy attains the same performance as the average of the performance of the permutations of P_π^σ of any given policy P_π . Since all permutations achieve the same performance, for any $P_\pi \in L^N$, there exists an exchangeable policy in L_{EX}^N that achieves the same performance.

Next, we show that we cannot strengthen the exchangeability condition to a symmetry condition without loss (in the finitely many sensor regime). We expand the example provided in [21], showing first that symmetric optimal policies are not optimal.

Example 1. Suppose that $N = 2$, $\mathbb{Y} = \{1, 2, 3\}$, and $\mathbb{U} = \{1, 2\}$. Also, let $P(H_1) = P(H_2) = 0.5$. Consider the following conditional distribution for observations: $P(y^i = 1 | H_1) = \frac{4}{5}$, $P(y^i = 2 | H_1) = \frac{1}{5}$, $P(y^i = 3 | H_1) = 0$, $P(y^i = 1 | H_2) = \frac{1}{3}$, $P(y^i = 2 | H_2) = \frac{1}{3}$, $P(y^i = 3 | H_2) = \frac{1}{3}$, for $i \in \{1, 2\}$. Restricting the search for an optimal policy to threshold policies, we get only two options: A) $u^i = 1$ if and only if $l^i = \frac{5}{12}$ (that is $y^i = 1$); B) $u^i = 1$ if and only if $l^i = \frac{5}{12}$ or $\frac{5}{3}$ (that is $y^i = 1$ or 2).

Suppose that sensors apply asymmetric policies first (for example, sensor 1 applies policy A while sensor 2 applies policy B). Then, the optimal response of the fusion center is $u^0 = 1$ if and only if $\Delta_2(u^1, u^2) > 0$ (which is equivalent to $u^0 = 1$ if and only if $u^1 = u^2 = 1$). The optimal cost is $J(\gamma^{0*:2*}) = 0.21$. Now, suppose that both sensors apply A. Then, an analogous analysis yields the cost $J(\gamma^{0*:2*}) = \frac{53}{225} = 0.23$. On the other hand, if they both apply policy B, then we get $J(\gamma^{0*:2*}) = \frac{2}{9} = 0.22$. This shows that asymmetric threshold policies can outperform symmetric ones for the finite population setting. Note that the symmetric independent sensor policy that assigns probability 0.5 to $\gamma^i = A$ and $\gamma^i = B$ for each sensor is also suboptimal with the total expected cost $\frac{1}{4}(\frac{2}{9} + \frac{53}{225} + \frac{38}{90}) = 0.22$. However, an exchangeable policy that assigns a probability of 0.5 to $(\gamma^1 = A, \gamma^2 = B)$ and $(\gamma^1 = B, \gamma^2 = A)$ is also optimal since the fusion center observes $\mu^2 = [1 \ 1]$ and given the indices of the sensor ($I_1 = 1, I_2 = 2$) or ($I_1 = 2, I_2 = 1$) can infer the realized policy is A, B or B, A, and hence, the performance is again 0.21. Hence, an exchangeable policy exists, but not a symmetric and independent.

We note that if we consider a Bayesian formulation, where the fusion center only observes $u^{1:N}$ and P_π , then again under the exchangeable policy that assigns a probability of 0.5 to $(\gamma^1 = A, \gamma^2 = B)$ and $(\gamma^1 = B, \gamma^2 = A)$, the optimal response of the fusion center is $u^0 = 1$ if and only if $u^1 = u^2 = 1$. Hence, the performance remains the same.

Although the same performance is achieved under the Bayesian fusion center in this example, this does not hold true in general. For example, suppose that both sensors observe the true hypothesis. Sensor 1 selects a policy C such that $P(u^1 = 1 | H_1) = 1$ and $P(u^1 = 2 | H_2) = 1$ while Sensor 2 selects a policy D such that $P(u^1 = 1 | H_1) = P(u^1 = 2 | H_1) = 0.5$ and $P(u^1 = 1 | H_2) = P(u^1 = 2 | H_2) = 0.5$. We observe that in the non-Bayesian setting, $\Delta(1, 1) = \Delta(1, 2) = \infty$ and $\Delta(2, 2) = \Delta(2, 1) = -\infty$. Hence, the probability of the error becomes 0, and hence, the optimal performance is 0. The permutation of this policy remains optimal as it again attains 0. However, in the Bayesian case under the exchangeable policy that assigns a probability of 0.5 to $(\gamma^1 = C, \gamma^2 = D)$ and $(\gamma^1 = D, \gamma^2 = C)$, we get $\Delta(2, 1) = \Delta(1, 2) = 0$, and hence, the probability of error becomes 0.25, clearly not optimal. Hence, an exchangeable policy might not be optimal in the Bayesian setting.

5. OPTIMALITY OF A SYMMETRIC POLICY FOR INFINITE POPULATION PROBLEM

In this section, we show that an optimal policy for the decentralized detection problem with infinitely many sensors is symmetric, independent, and of threshold type. Now, we establish the existence of an identical (possibly randomized) encoding policy for the sensors.

Theorem 3. Consider $\mathcal{P}_{\text{E}}^N$ and $\mathcal{P}_{\text{E}}^\infty$. Let Assumption 1 hold and for $j = 1, 2$, and $i \geq 1$, $\mathbb{E}[(\log l^i)^2 | H_j] < \infty$. Then,

- (i) There exists an optimal solution $(\gamma^{0*}, \gamma^{R*}, \gamma^{R*}, \dots)$ for $\mathcal{P}_{\text{E}}^\infty$ that is a MAP rule for the fusion center and independent (possibly randomized) and also identical policy for each sensor.
- (ii) An identical optimal solution $(\gamma^{0*}, \gamma^{R*}, \gamma^{R*}, \dots)$ for $\mathcal{P}_{\text{E}}^\infty$ restricted to N -sensors is ϵ_N -optimal for $\mathcal{P}_{\text{E}}^N$ with $\epsilon_N \rightarrow 0$ as $N \rightarrow \infty$.

We now summarize the main steps of our proof technique. We first use the lower bound for the expected cost J_{EE}^N for any encoding policy from [21, Lemma 1] and [30]. Then, we show that the limsup of the optimal expected cost J_{EE}^N under exchangeable encoding policies is the same as the limsup of that under infinitely exchangeable policies. We note that Theorem 2 only allows us to restrict encoding policies to finite exchangeable policies (which might not be identical and independent). Here, we use a probabilistic argument based on a finite de Finetti theorem and a convergence result of an exchangeable sequence of random variables by Aldous [31] to show that the restriction can be refined to infinitely exchangeable encoding policies. This allows us to use a de Finetti theorem [31] to show the optimality of identical and independent sensors encoding policies.

6. CONCLUSION

We studied a class of decentralized binary detection problems involving a single fusion center and many sensors. In the finite population setting, we established the existence and optimality of an independent threshold-based encoding policy for the sensors and a MAP rule for the fusion center. Additionally, we showed that an exchangeable encoding policy is also optimal. In the infinite population setting, we proved the existence and optimality of an independent, symmetric encoding policy under the error exponent cost.

We anticipate that the mathematical program used in the paper, building on [22], will find applications in several other massive communications applications [32, 33].

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